Re-analysis Summary Report for Peer-evaluation

Paper ID: **Anderson_AmEcoJourn_2011_bLe8**

Re-analyst's ID: **jru4j**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
… [the study] finds substantially higher income for low-caste households residing in villages dominated by BACs [compared to villages dominated by upper caste]. (p. 240.)

The corresponding article

Here you can find the original article: https://osf.io/hcxdq/files/osfstorage/5e56aac535d22d02f75f482c?view_only=32ccf80d2c684ea9ac10ebbc4ec34df6
and the original data: https://osf.io/k7xzy/?view_only=c1f2b9aaf05d486c942783efd8ec57ec

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
The analysis compares the income of lower caste households as a function of whether their village was upper caste dominated or lower caste dominated. We expect that households in villages with lower caste dominance will have greater income compared to those who live in villages with higher caste dominance. A linear mixed model (estimated using REML and nloptwrap optimizer) was fitted to predict total income of households with caste membership, the dominant population, and their interaction as predictors. The model included the village the household lived as random effect. The model to test the hypothesis was defined like this to control for the effect of villages being different in many aspects other than their dominant caste, and to see whether different castes fare worse compared to others in different dominance settings. The model's total explanatory power is substantial (conditional $R2 = 0.26$) and the part related to the fixed effects alone (marginal R2) is 0.09. The model's intercept, corresponding to caste 3 and lower caste dominance, is at 11768.02 (95% CI [9678.09, 13857.94], $t(1287) = 11.05$, $p < .001$). Within this model: The effect of being a member of caste 4 compared to caste 3 is statistically significant and negative ($b = -6858.49$, 95% CI [-9245.47, -4471.52], $t(1287) = -5.64$, $p < .001$; Std. beta = -0.53, 95% CI [-0.72, -0.35]). The effect of being

a member of caste 5 compared to caste 3 is statistically significant and negative (b = -7612.83, 95% CI [-9683.41, -5542.25], t(1287) = -7.21, p < .001; Std. beta = -0.59, 95% CI [-0.75, -0.43]). The effect of village being a high caste dominated village compared to it being low caste dominated is statistically significant and negative (beta = -4753.76, 95% CI [-7983.86, -1523.66], t(1287) = -2.89, p = 0.004; Std. beta = -0.37, 95% CI [-0.62, -0.12]). The interaction effect of high caste dominance on being a member of caste 4 compared to caste 3 is statistically significant and positive (beta = 3682.35, 95% CI [90.32, 7274.39], t(1287) = 2.01, p = 0.045; Std. beta = 0.29, 95% CI [7.03e-03, 0.57]). The interaction effect of high caste dominance on being a member of caste 5 is statistically non-significant and positive (beta = 1993.58, 95% CI [-1123.48, 5110.64], t(1287) = 1.25, p = 0.210; Std. beta = 0.16, 95% CI [-0.09, 0.40]). Standardized parameters were obtained by fitting the model on a standardized version of the dataset.

The analyst's conclusion: Based on the results, we see that households that live in high cast dominant villages earn less, that this effect is different between castes, and that caste dominance has an effect on the effect of being a member of lower castes.

The analyst's categorization of the result:
The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
You should not use District controls, Crop controls, Distance controls, Groundwater controls, Public goods controls Do not include Irrigation, credit, and tenancy variables, and their interaction term versions Do not use instruments for water buyer status.

Here, the analyst reported the following steps and results of the analysis:
The analysis compares the income of lower caste households as a function of whether their village was upper caste dominated or lower caste dominated. We expect that households in villages with lower caste dominance will have greater income compared to those who live in villages with higher caste dominance. A linear mixed model (estimated using REML and nloptwrap optimizer) was fitted to predict total income of households with caste membership, the dominant population, and their interaction as predictors. The model included the village the household lived as random effect. The model to test the hypothesis was defined like this to control for the effect of villages being different in many aspects other than their dominant caste, and to see whether different castes fare worse compared to others in different dominance settings. The model's total explanatory power is substantial (conditional R2 = 0.26) and the part related to the fixed effects alone (marginal R2) is 0.09. In order to satisfy the instructions given in Task 2, I included the following analysis: The omnibus F-test of the effect of dominance on income is significant at the = 0.05 level, F (1, 93.8) = 4.441, p = 0.038. The model's intercept, corresponding to caste 3 and lower caste dominance, is at 11768.02 (95% CI [9678.09, 13857.94], t(1287) = 11.05, p < .001). Within this model: The effect of being a member of caste 4 compared to caste 3 is statistically significant and negative (b = -6858.49, 95% CI [-9245.47, -4471.52], t(1287) = -5.64, p < .001; Std. beta = -0.53, 95% CI [-0.72, -0.35]). The effect of being a member of caste 5 compared to caste 3 is statistically significant and negative (b = -7612.83, 95% CI [-9683.41, -5542.25], t(1287) = -7.21, p < .001; Std. beta = -0.59, 95% CI [-0.75, -0.43]). The effect of village being a high caste dominated village compared to it being low caste dominated is statistically significant and negative (beta = -4753.76, 95% CI [-7983.86, -1523.66], t(1287) = -2.89, p = 0.004; Std. beta = -0.37, 95% CI [-0.62, -0.12]). The interaction effect of high caste dominance on being a member of caste 4 compared to caste 3 is statistically significant and positive (beta = 3682.35, 95% CI [90.32, 7274.39], t(1287) = 2.01, p = 0.045; Std. beta = 0.29, 95% CI [.007, 0.57]). The interaction effect of high caste dominance on being a member of caste 5 is statistically non-significant and positive (beta = 1993.58, 95% CI [-1123.48, 5110.64], t(1287) = 1.25, p = 0.210; Std. beta = 0.16, 95% CI [-0.09, 0.40]). Standardized parameters were obtained by fitting the model on a standardized version of the dataset.

Analysis language/software/code (optional to check):
Task 1: R
Task 2: R
Analysis codes: https://osf.io/vmf7q/